Date: 29/10/26

Task 2 : Data Analysis , Aggregate functions, Data Comparison and Filtering,Charts ,Insights

Create Dataset:

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

```
data = {
    "OrderID": [1001,1002,1003,1004,1005,1006,1007,1008,1009,1010,1011,1012,1013,1014,1015,1016,1017,1018,1019,1020],
    "OrderDate": ["2025-01-05","2025-01-08","2025-01-12","2025-01-18","2025-02-03","2025-02-07","2025-02-14","2025-02-21","2025-03-02","2025-03-09","2
    "Region": ["North","South","East","West","South","North","East","West","North","South","East","West","North","South","East","West","North","South"
    "Category": ["Electronics","Furniture","Office Supplies","Electronics","Electronics","Furniture","Electronics","Office Supplies","Electronics","Fu
    "Product": ["Laptop","Chair","Printer Paper","Headphones","Monitor","Desk","Keyboard","Notebook","Mouse","Bookshelf","Stapler","Tablet","Office Ch
    "Quantity": [2,5,10,4,3,2,6,20,8,3,12,2,4,3,25,2,5,15,2,3],
    "UnitPrice": [65000,4500,550,2200,14500,9000,1800,180,900,7200,350,28000,6500,24000,120,11000,6500,250,42000,8500],
    "Sales": [130000,22500,5500,8800,43500,18000,10800,3600,7200,21600,4200,56000,26000,72000,3000,22000,32500,3750,84000,25500],
    "Discount": [5,10,0,5,8,5,0,2,5,10,0,7,5,10,0,5,8,2,6,10],
    "Profit": [18000,5000,1200,1800,6500,3200,2200,900,1500,3600,900,8500,5200,11000,700,4000,5500,800,12500,4300]
}

df = pd.DataFrame(data)
df.to_csv("sales_dataset_20_rows.csv", index=False)
df.head()
```

	OrderID	OrderDate	Region	Category	Product	Quantity	UnitPrice	Sales	Discount	Profit
0	1001	2025-01-05	North	Electronics	Laptop	2	65000	130000	5	18000
1	1002	2025-01-08	South	Furniture	Chair	5	4500	22500	10	5000
2	1003	2025-01-12	East	Office Supplies	Printer Paper	10	550	5500	0	1200
3	1004	2025-01-18	West	Electronics	Headphones	4	2200	8800	5	1800
4	1005	2025-02-03	South	Electronics	Monitor	3	14500	43500	8	6500

Basic info:

```
df.shape
```

```
(20, 10)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 10 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderID    20 non-null     int64
 1   OrderDate  20 non-null     str
 2   Region     20 non-null     str
 3   Category   20 non-null     str
 4   Product    20 non-null     str
 5   Quantity   20 non-null     int64
 6   UnitPrice  20 non-null     int64
 7   Sales      20 non-null     int64
 8   Discount   20 non-null     int64
 9   Profit     20 non-null     int64
dtypes: int64(6), str(4)
memory usage: 1.7 KB
```

```
df.describe()
```

	OrderID	Quantity	UnitPrice	Sales	Discount	Profit
count	20.00000	20.000000	20.00000	20.00000	20.000000	20.000000
mean	1010.50000	6.800000	11652.50000	30022.50000	5.150000	4865.000000
std	5.91608	6.517991	16676.32923	32954.81945	3.558163	4570.416888
min	1001.00000	2.000000	120.00000	3000.00000	0.000000	700.000000
25%	1005.75000	2.750000	812.50000	6775.00000	2.000000	1425.000000
50%	1010.50000	4.000000	6500.00000	21800.00000	5.000000	3800.000000
75%	1015.25000	8.500000	11875.00000	35250.00000	8.000000	5750.000000
max	1020.00000	25.000000	65000.00000	130000.00000	10.000000	18000.000000

Aggregate Functions:

```
df["Sales"].sum()
```

```
np.int64(600450)
```

```
df["Sales"].mean()
```

```
np.float64(30022.5)
```

```
df["Sales"].min()
```

```
np.int64(3000)
```

```
df["Sales"].max()
```

```
np.int64(130000)
```

```
df["Sales"].count()
```

```
np.int64(20)
```

```
df.groupby("Region")["Sales"].sum().sort_values(ascending=False)
```

```
Region
North    213700
South    163350
West     115900
East     107500
Name: Sales, dtype: int64
```

```
df.groupby("Region")["Sales"].agg(["sum","mean","min","max","count"])
```

Region	sum	mean	min	max	count
East	107500	21500.0	3000	84000	5
North	213700	42740.0	7200	130000	5
South	163350	32670.0	3750	72000	5
West	115900	23180.0	3600	56000	5

```
df.groupby("Category").agg({"Sales":["sum","mean"],"Quantity":"sum","Profit":"sum"}).sort_values(("Sales","sum"), ascending=False)
```

	Sales		Quantity	Profit
	sum	mean	sum	sum
Category				
Electronics	444800	49422.222222	35	67500
Furniture	135600	22600.000000	19	25300
Office Supplies	20050	4010.000000	82	4500

New Column:

```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
df["Profit_Margin"] = (df["Profit"] / df["Sales"]) * 100
df["Sales_Per_Unit"] = df["Sales"] / df["Quantity"]
df.head()
```

	OrderID	OrderDate	Region	Category	Product	Quantity	UnitPrice	Sales	Discount	Profit	Revenue	Profit_Margin	Sales_Per_Unit
0	1001	2025-01-05	North	Electronics	Laptop	2	65000	130000	5	18000	130000	13.846154	65000.0
1	1002	2025-01-08	South	Furniture	Chair	5	4500	22500	10	5000	22500	22.222222	4500.0
2	1003	2025-01-12	East	Office Supplies	Printer Paper	10	550	5500	0	1200	5500	21.818182	550.0
3	1004	2025-01-18	West	Electronics	Headphones	4	2200	8800	5	1800	8800	20.454545	2200.0
4	1005	2025-02-03	South	Electronics	Monitor	3	14500	43500	8	6500	43500	14.942529	14500.0

```
df[["OrderID","Quantity","UnitPrice","Sales","Revenue","Profit_Margin","Sales_Per_Unit"]]
```

	OrderID	Quantity	UnitPrice	Sales	Revenue	Profit_Margin	Sales_Per_Unit
0	1001	2	65000	130000	130000	13.846154	65000.0
1	1002	5	4500	22500	22500	22.222222	4500.0
2	1003	10	550	5500	5500	21.818182	550.0
3	1004	4	2200	8800	8800	20.454545	2200.0
4	1005	3	14500	43500	43500	14.942529	14500.0
5	1006	2	9000	18000	18000	17.777778	9000.0
6	1007	6	1800	10800	10800	20.370370	1800.0
7	1008	20	180	3600	3600	25.000000	180.0

Data Comparison and filtering:

[16]:
```
df.sort_values("Sales", ascending=False)[["OrderID","Region","Category","Product","Sales","Profit"]].head(5)
```

[16]:

	OrderID	Region	Category	Product	Sales	Profit
0	1001	North	Electronics	Laptop	130000	18000
18	1019	East	Electronics	Camera	84000	12500
13	1014	South	Electronics	Smartphone	72000	11000
11	1012	West	Electronics	Tablet	56000	8500
4	1005	South	Electronics	Monitor	43500	6500

[17]:
```
df[df["Sales"] > df["Sales"].mean()][["OrderID","Region","Product","Sales"]]
```

[17]:

	OrderID	Region	Product	Sales
0	1001	North	Laptop	130000
4	1005	South	Monitor	43500
11	1012	West	Tablet	56000
13	1014	South	Smartphone	72000
16	1017	North	Smartwatch	32500
18	1019	East	Camera	84000

:
```
df[df["Profit"] == df["Profit"].max()][["OrderID","Product","Sales","Profit"]]
```

:

	OrderID	Product	Sales	Profit
0	1001	Laptop	130000	18000

:
```
df.groupby("Product")["Sales"].sum().sort_values(ascending=False).head(10)
```

:
```
Product
Laptop          130000
Camera           84000
Smartphone       72000
Tablet           56000
Monitor          43500
Smartwatch       32500
Office Chair     26000
Cabinet          25500
Chair            22500
Table            22000
Name: Sales, dtype: int64
```

:
```
df.groupby("Region")["Profit"].sum().sort_values(ascending=False)
```

:
```
Region
North    33400
South    26900
West     19500
East     17500
Name: Profit, dtype: int64
```

Date-Time conversions:

[22]:
```
df["OrderDate"] = pd.to_datetime(df["OrderDate"])
df.dtypes
```

[22]:
```
OrderID                    int64
OrderDate         datetime64[us]
Region                       str
Category                     str
Product                      str
Quantity                   int64
UnitPrice                  int64
Sales                      int64
Discount                   int64
Profit                     int64
Revenue                    int64
Profit_Margin            float64
Sales_Per_Unit           float64
dtype: object
```

[23]:
```
df["Year"] = df["OrderDate"].dt.year
df["Month"] = df["OrderDate"].dt.month
df["Month_Name"] = df["OrderDate"].dt.month_name()
df["Day"] = df["OrderDate"].dt.day
df["Day_Name"] = df["OrderDate"].dt.day_name()

df[["OrderID","OrderDate","Year","Month","Month_Name","Day","Day_Name"]]
```

[23]:

	OrderID	OrderDate	Year	Month	Month_Name	Day	Day_Name
0	1001	2025-01-05	2025	1	January	5	Sunday
1	1002	2025-01-08	2025	1	January	8	Wednesday
2	1003	2025-01-12	2025	1	January	12	Sunday
3	1004	2025-01-18	2025	1	January	18	Saturday

Date based Analysis:

[24]:
```
df.groupby("Year")["Sales"].sum()
```

[24]:
```
Year
2025    600450
Name: Sales, dtype: int64
```

[25]:
```
df.groupby(["Year","Month"])["Sales"].sum()
```

[25]:
```
Year  Month
2025  1        166800
      2         75900
      3         89000
      4        101000
      5         58250
      6        109500
Name: Sales, dtype: int64
```

[26]:
```
df.groupby("Month_Name")["Sales"].sum().reindex(df["Month_Name"].drop_duplicates())
```

[26]:
```
Month_Name
January     166800
February     75900
March        89000
April       101000
May          58250
June        109500
Name: Sales, dtype: int64
```

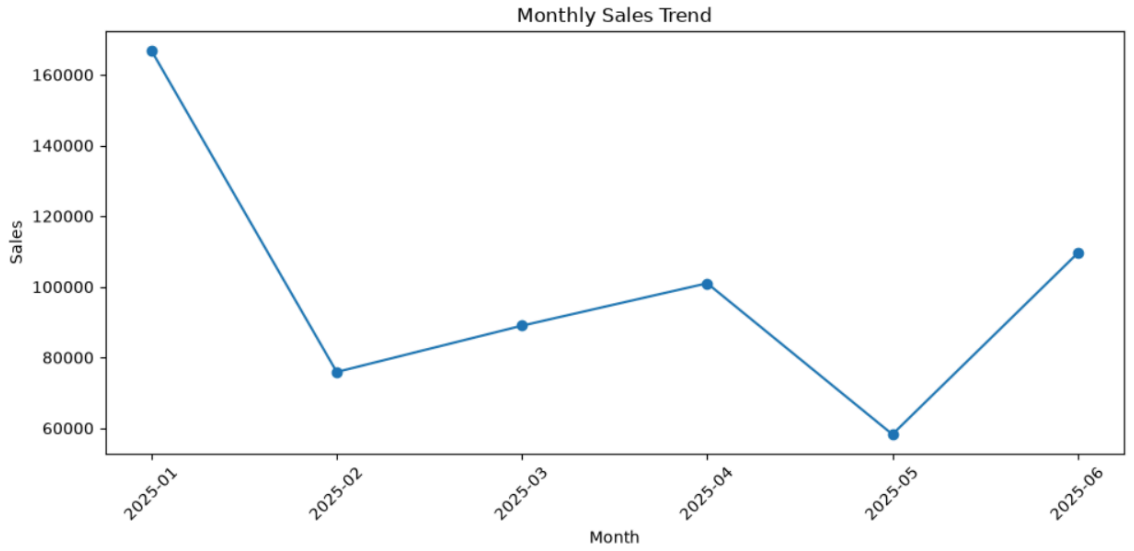
[27]:
```
df.groupby("Day_Name")["Sales"].sum()
```

[27]:
```
Day_Name
Friday       130400
Monday        43500
Saturday      72000
Sunday       164300
Tuesday      167750
```

Charts:

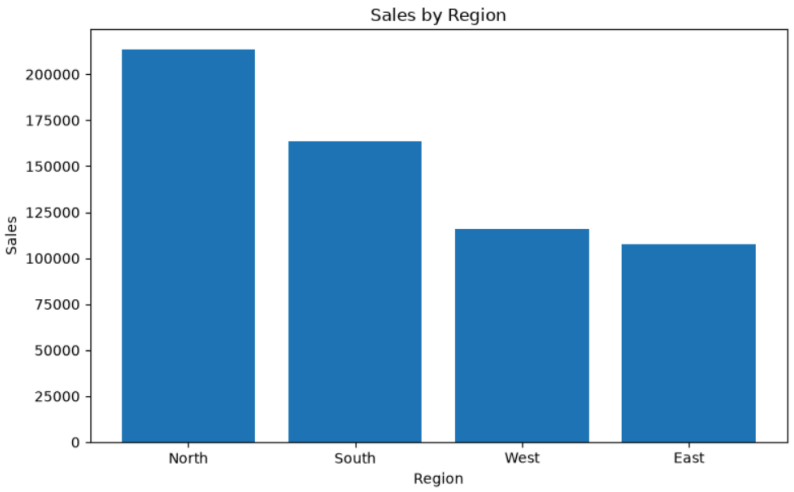
```
monthly_sales = df.groupby(df["OrderDate"].dt.to_period("M"))["Sales"].sum()

plt.figure(figsize=(10,5))
plt.plot(monthly_sales.index.astype(str), monthly_sales.values, marker="o")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales Trend")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



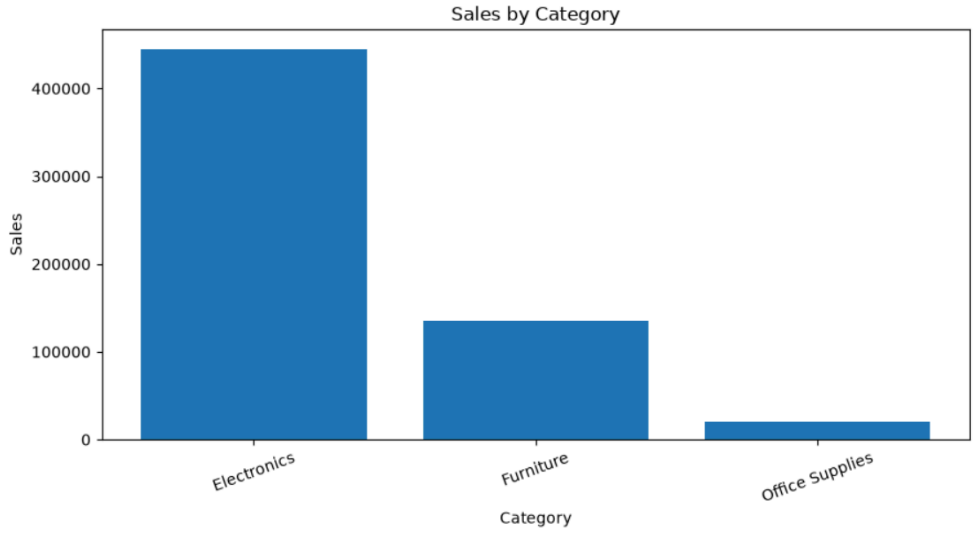

```
region_sales = df.groupby("Region")["Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(8,5))
plt.bar(region_sales.index, region_sales.values)
plt.xlabel("Region")
plt.ylabel("Sales")
plt.title("Sales by Region")
plt.tight_layout()
plt.show()
```



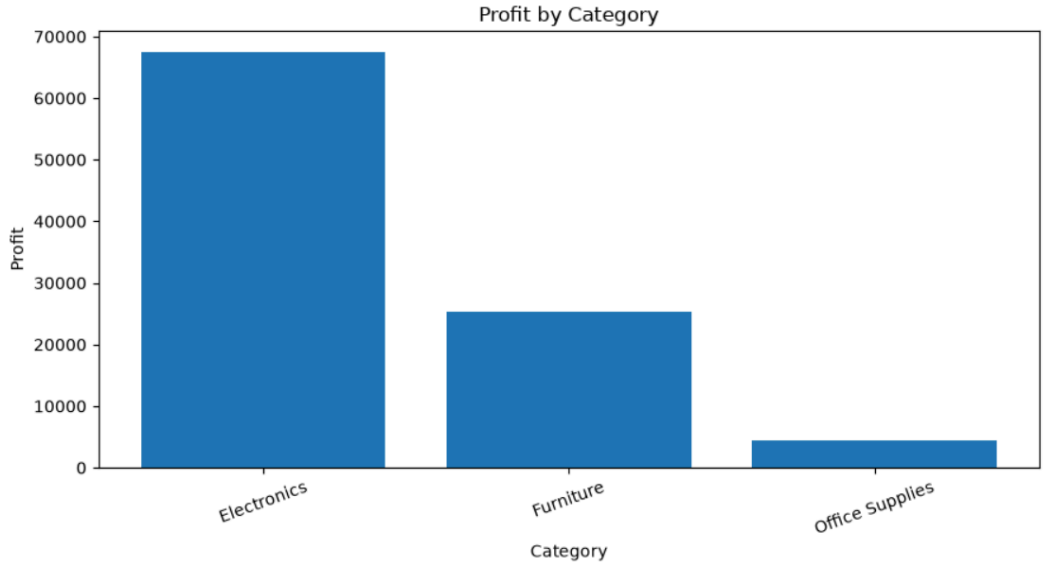
```
category_sales = df.groupby("Category")["Sales"].sum().sort_values(ascending=False)

plt.figure(figsize=(9,5))
plt.bar(category_sales.index, category_sales.values)
plt.xlabel("Category")
plt.ylabel("Sales")
plt.title("Sales by Category")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

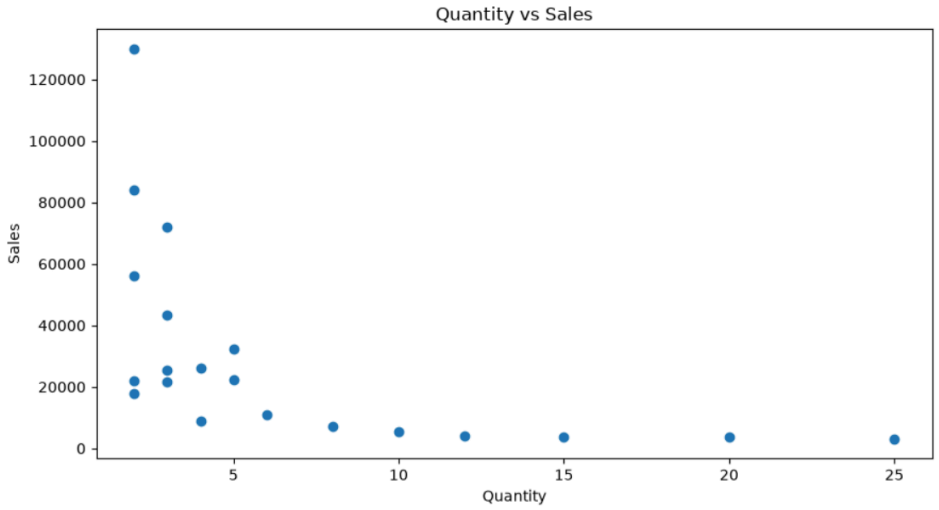



```
category_profit = df.groupby("Category")["Profit"].sum().sort_values(ascending=False)

plt.figure(figsize=(9,5))
plt.bar(category_profit.index, category_profit.values)
plt.xlabel("Category")
plt.ylabel("Profit")
plt.title("Profit by Category")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(9,5))
plt.scatter(df["Quantity"], df["Sales"])
plt.xlabel("Quantity")
plt.ylabel("Sales")
plt.title("Quantity vs Sales")
plt.tight_layout()
plt.show()
```



Key Results ¶

```
top_region = df.groupby("Region")["Sales"].sum().idxmax()
top_region_sales = df.groupby("Region")["Sales"].sum().max()

top_category = df.groupby("Category")["Sales"].sum().idxmax()
top_category_sales = df.groupby("Category")["Sales"].sum().max()

top_product = df.groupby("Product")["Sales"].sum().idxmax()
top_product_sales = df.groupby("Product")["Sales"].sum().max()

best_profit_product = df.loc[df["Profit"].idxmax(), "Product"]
best_profit = df["Profit"].max()

highest_month = df.groupby(df["OrderDate"].dt.to_period("M"))["Sales"].sum().idxmax()
highest_month_sales = df.groupby(df["OrderDate"].dt.to_period("M"))["Sales"].sum().max()

print("Total Sales:", df["Sales"].sum())
print("Average Sales:", df["Sales"].mean())
print("Highest Sales:", df["Sales"].max())
print("Lowest Sales:", df["Sales"].min())
print("Total Profit:", df["Profit"].sum())
print("Top Region:", top_region, "-", top_region_sales)
print("Top Category:", top_category, "-", top_category_sales)
print("Top Product:", top_product, "-", top_product_sales)
print("Highest Profit Product:", best_profit_product, "-", best_profit)
print("Best Sales Month:", highest_month, "-", highest_month_sales)
```

```
Total Sales: 600450
Average Sales: 30022.5
Highest Sales: 130000
Lowest Sales: 3000
Total Profit: 97300
Top Region: North - 213700
Top Category: Electronics - 444800
Top Product: Laptop - 130000
Highest Profit Product: Laptop - 18000
Best Sales Month: 2025-01 - 166800
```

```python
insights = [
    f"{top_region} has the highest total sales.",
    f"{top_category} is the top-performing category by sales.",
    f"{top_product} is the highest-selling product.",
    f"{best_profit_product} generates the highest individual profit.",
    f"{highest_month} has the highest monthly sales.",
    f"The average order sales value is {df['Sales'].mean():.2f}.",
    f"The total profit across all orders is {df['Profit'].sum():.2f}."
]

for i, insight in enumerate(insights, 1):
    print(f"{i}. {insight}")
```

```
1. North has the highest total sales.
2. Electronics is the top-performing category by sales.
3. Laptop is the highest-selling product.
4. Laptop generates the highest individual profit.
5. 2025-01 has the highest monthly sales.
6. The average order sales value is 30022.50.
7. The total profit across all orders is 97300.00.
```